Bingxuan Li

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Research Interests

My research lies at the intersection of deep learning and computer graphics, with a focus on efficient visual representations and 3D visual understanding. My recent work spans physics-informed monocular depth estimation and Gaussian representations.

Education

New York University — New York City, USA
Doctor of Philosophy in Computer Science — *Sep 2024 – Present*
Advisor: Prof. Qi Sun

Peking University — Beijing, China
Bachelor of Science in Computer Science with Honours, Turing Class — *Sep 2020 – Jun 2024*

Work Experience

Research Intern, TikTok — San Jose, USA
Improved motion generation quality through post-training techniques. — *May 2026 – Aug 2026*
Mentors: Dr. Yili Zhao, Dr. Siqi Wang, Xinhao Song

Research

- **Physically Grounded Monocular Depth via Nanophotonic Wavefront Encoding**
  ECCV 2026 | Paper | Project | Code | Video
  Bingxuan Li*, Jiahao Wu*, Yuan Xu*, Zezheng Zhu, Yunxiang Zhang, Kenneth Chen, Yanqi Liang, Nanfang Yu†, Qi Sun†
- **Image-GS: Content-Adaptive Image Representation via 2D Gaussians**
  ACM SIGGRAPH 2025 | Paper | Project | Video
  Yunxiang Zhang*, **Bingxuan Li***, Alexandr Kuznetsov, Akshay Jindal, Kenneth Chen, Anton Sochenov, Anton Kaplanyan, Qi Sun†
- **Proxy Tracing: Unbiased Reciprocal Estimation for Optimized Sampling in BDPT**
  ACM Transactions on Graphics (ACM SIGGRAPH 2024) | Paper | Project | Video
  Fujia Su*, **Bingxuan Li***, Qingyang Yin, Yanchen Zhang, Sheng Li†
- **Nano-Optics for Depth Sensing**
  ACM SIGGRAPH 2025 Emerging Technologies | Paper
  Yuan Xu*, **Bingxuan Li***, Jiahao Wu*, Yunxiang Zhang, Zezheng Zhu, Nanfang Yu†, Qi Sun†
- **Frequency-Adaptive Gaussian Grids for Efficient Image Representation**
  WACV 2027
  Anatolii Evdokimov, Stavros Diolatzis, **Bingxuan Li**, Akshay Jindal, Patsorn Sangkloy, Qi Sun†

* Equal contributions † Corresponding authors

Awards

New York University — SoE Fellowship (2024)
Peking University — John Hopcroft Scholarship (2023)
Peking University — John Hopcroft Scholarship (2021)
Peking University — Freshman Scholarship (2020)
Chinese Chemistry Olympiad — Silver Medal (2018)

Skills

- **Programming**: Python, C/C++
- **Tools**: PyTorch, CUDA
- **Language**: Mandarin, English